

Does Data Scaling Lead to Visual Compositional Generalization?

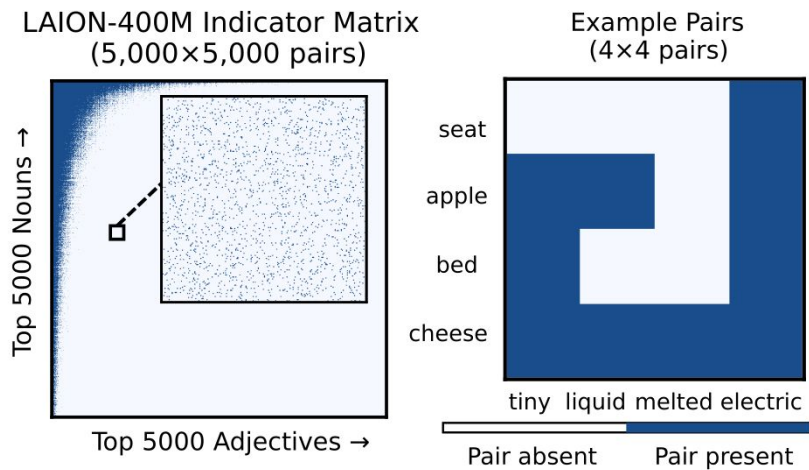
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Introduction

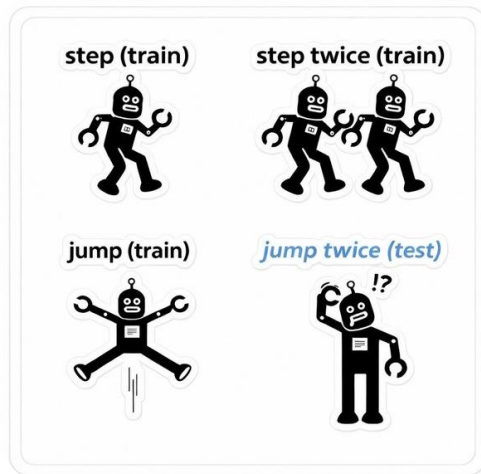
Motivation

- Humans generalize by **combining known concepts**
- Vision models still struggle with **novel concept combinations**



What is Compositional Generalization?

- Concepts = building blocks (color, shape, size, etc.)
- Compositional generalization = *Correctly recognizing unseen combinations of known concepts*
- Seen: red square, blue circle
- Unseen: red circle ✗ / ✓ ?



Central Research Question

Main question

Does scaling data alone lead to compositional generalization?

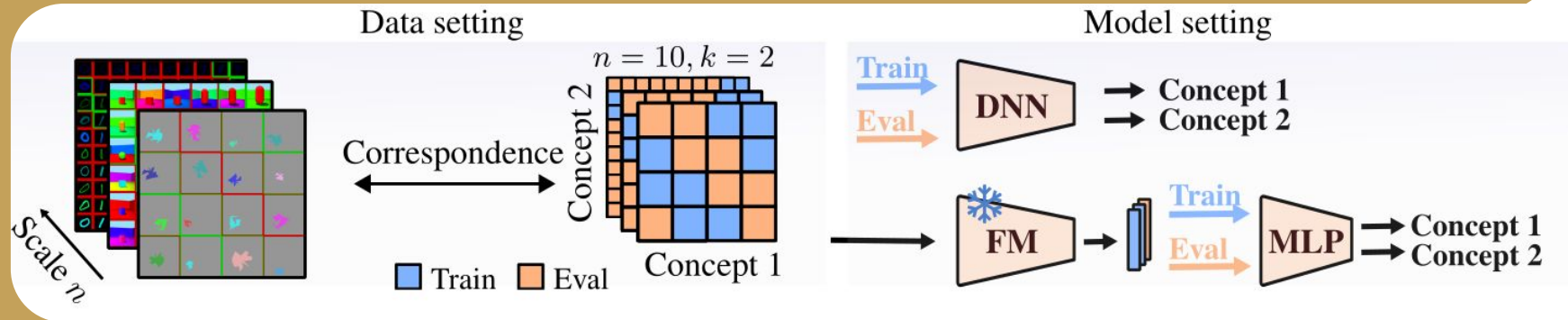
Sub-questions

- Is **more data** enough?
- Or do we need **more diverse combinations**?
- What kind of **representation structure** enables generalization?

One-liner

Scale vs. diversity vs. structure

Framework



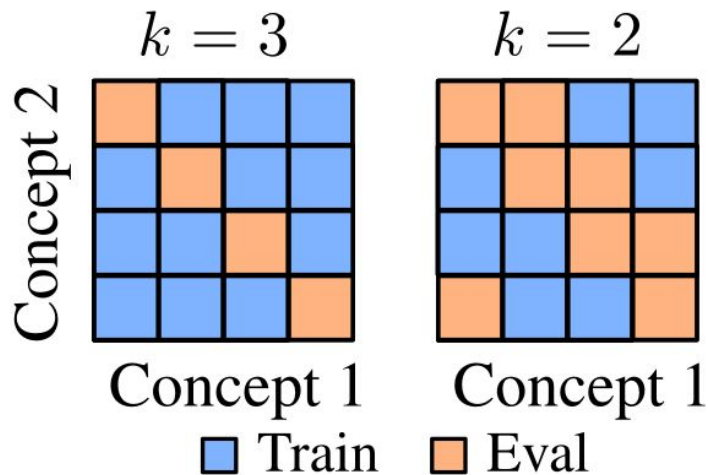
The (n, k) Experimental Framework

Framework

- Two concepts (e.g., color $\times$ shape)
- n = number of values per concept
- k = number of seen combinations per concept value

Example

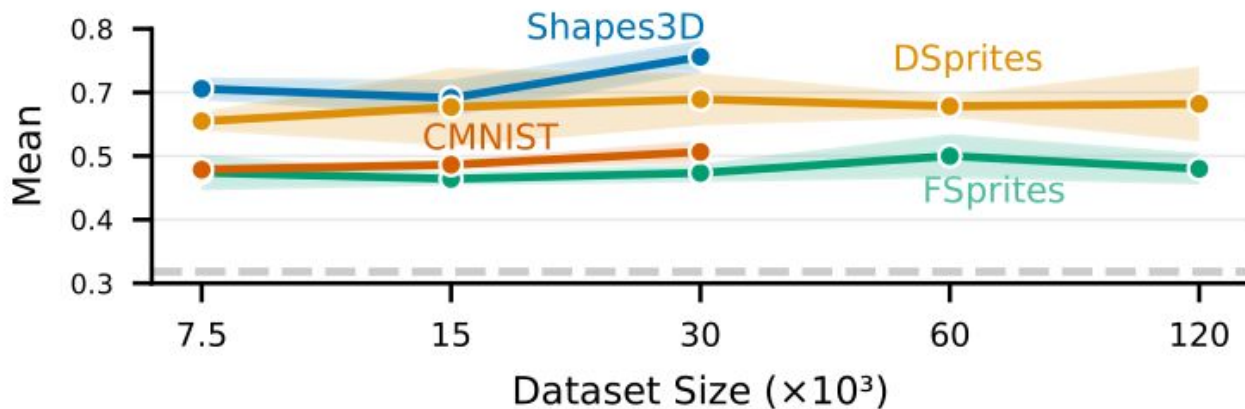
- $n = 4, k = 2$
- Only 8 training combinations out of 16 possible



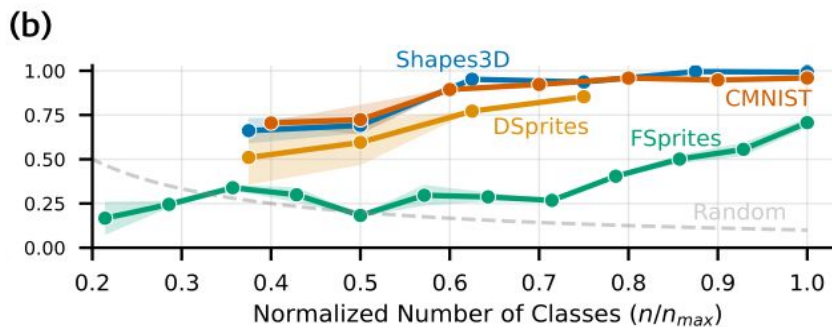
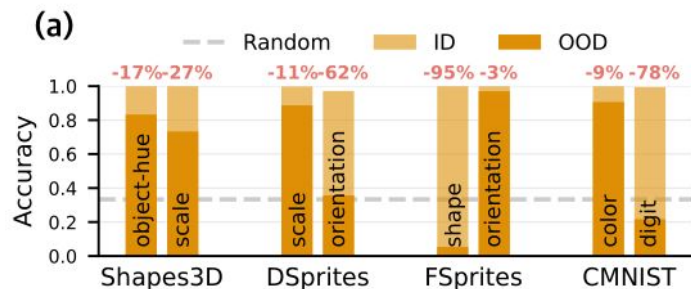
Does More Data Help?

Experiment

- Increase **in-distribution** dataset size
- Keep combinations fixed



Does More Diversity Help?

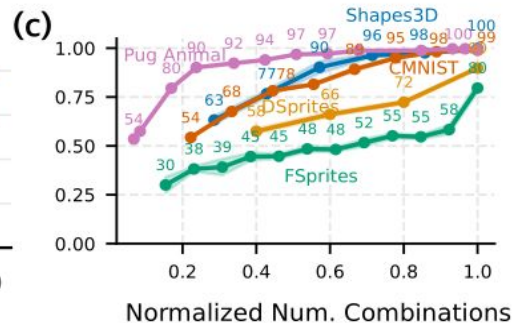


Two ways to increase diversity

1. Increase n (more concept values)
2. Increase k (more combinations)

Result

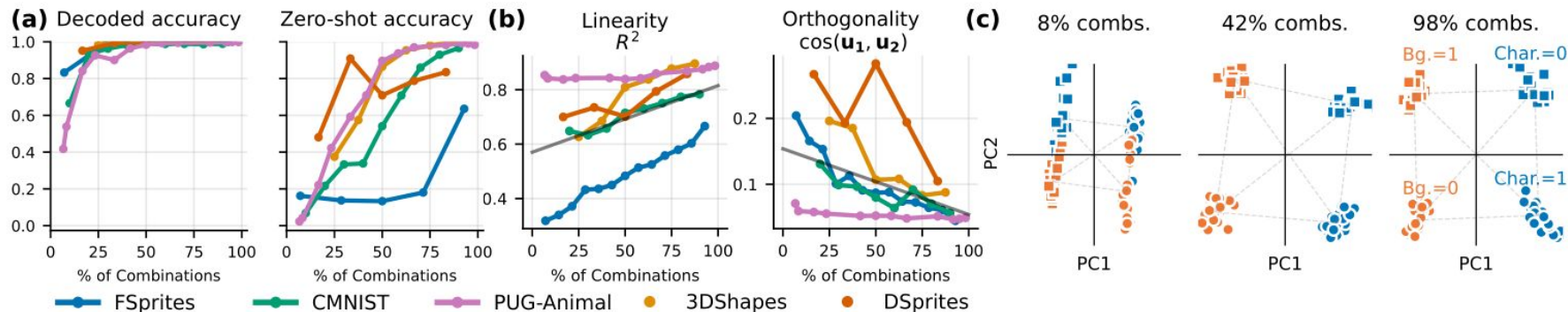
- Both significantly improve OOD accuracy



What Happens Inside the Model?

Models go through **three learning phases** as diversity increases:

- **Spurious features** (low diversity)
- **Discriminative but entangled features**
- **Linearly factored representations**



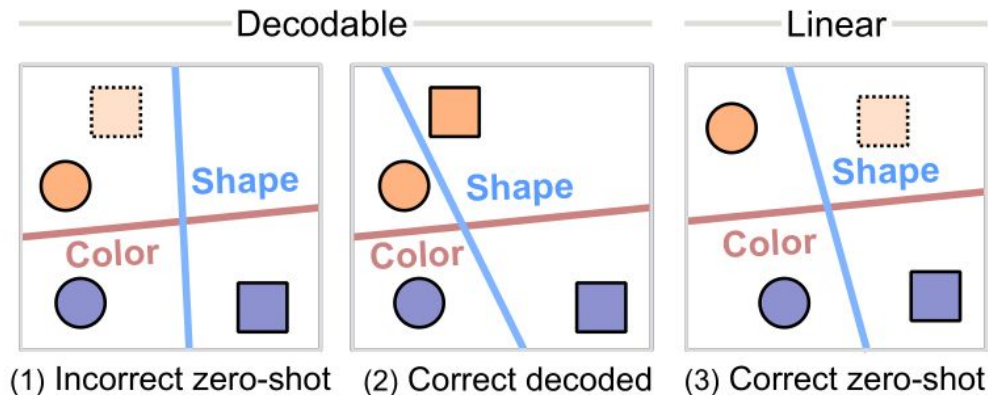
Linearly Factored Representations

Definition (intuitive)

- Representation of (color, shape) $\approx$ **color vector** + **shape vector**

Why it matters

- Enables systematic recombination
- Explains strong zero-shot performance



Theoretical Result

Proposition

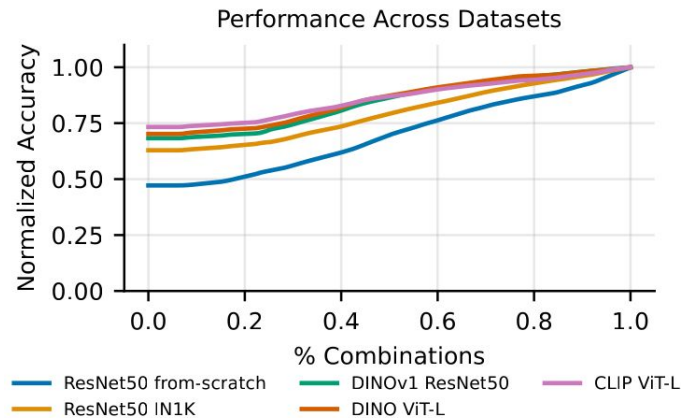
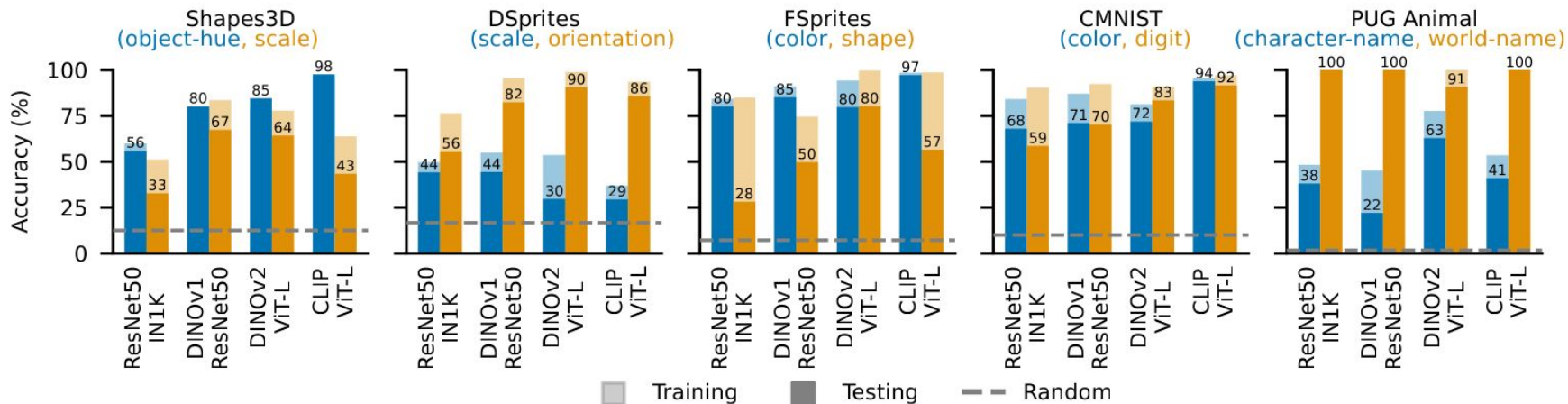
- If representations are linearly factored:
 - Only **$k = 2$ examples per concept value**
 - Enough for **perfect generalization**

Why it's powerful

- Explains data efficiency
- Connects geometry $\rightarrow$ generalization

Proposition 4.1 (Minimal Compositional Learning). *Let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ be a feature extractor with linearly factored concept embeddings over $\mathcal{C}$. Let $\{\mathbf{u}_{c_1^1}, \dots, \mathbf{u}_{c_1^n}\}$ and $\{\mathbf{u}_{c_2^1}, \dots, \mathbf{u}_{c_2^n}\}$ be the concept vectors for the first and second concepts respectively, where their joint span has dimension $2n - 1$. Suppose we only observe joint representations for concept combinations $c_i, c_j \in \{1, \dots, n\}$. Then $k = 2$ combinations per concept value suffice to learn a linear classifier that perfectly generalizes to all $(n - k) \cdot n$ unseen combinations.*

What About Pretrained Models?





Any Question?

Thank you for your attention!

